Introduction to Vv186

Wenxin Chen

Table of contents

1. Info About the Course

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Faculty Directory

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Education

Ph.D. (Dr. rer. nat.)	University of Potsdam, Germany (2006)
Physics Diploma ≈ M.S.	University of Potsdam (2001)
Chemistry pre-Diploma ≈ B.S.	University of Potsdam (1995)

Work Experience

2021.09 – pres.	Teaching Professor, UM-SJTU Joint Institute at the SJTU
2014.09 – 2021.08	Associate Teaching Professor, UM-SJTU Joint Institute at the SJTU
2007.03 – 2014.08	Lecturer, UM-SJTU Joint Institute at the SJTU
2001.10 – 2007.03	Research Assistant, University of Potsdam

Awards

- Shanghai Municipal Level Key Course award (for Vv286 Honors Mathematics IV)
- 2010 SJTU Outstanding Teacher Award: second class

The course will focus on functions of one variable, in particular:

- Foundations: Set Theory, Logic, Numbers, Algeraic Structures.
- Functions, Limits and Continuity,
- Differentiation
- Integration

Important Infos You Must Know:

- Lectures will be **ONLINE**, but I still recommend you to watch every course on time and engage actively.
- "Formally, our course is one of three parallel courses on functions of a single variable available to you; ... MATH1560J Honors Calculus II has a greater emphasis on applications and modeling."

2. What You Will Learn and How to Learn Vv186 Well

What You Will Learn

Lecture	Lecture Subject
1	Elements of Logic
2	Set Theory
3	Natural, Rational and Real Numbers
4	Complex Numbers
5	Sequences and Convergence
6	Sequences and Convergence
7	Metric Spaces, Real Functions
8	Limits and Landau Symbols
9	Properties of Continuous Functions
10	Properties of Continuous Functions
11	**Midterm Exam**
12	The Derivative
13	The Derivative
14	Properties of Differentiable Functions
15	Properties of Differentiable Functions
16	Vector Spaces
17	Sequences of Functions
18	Series
19	Series
20	Power Series
21	The Exponential and Logarithm Functions
22	The Trigonometric Functions
23	Step Functions and Regulated Functions
24	The Regulated and Darboux Integrals
25	The Riemann Integral
26	Integration in Practice
27	Integration in Practice
28	Applications of Integration
29	Applications of Integration
30	**Final Exam**

- Construct all the outcomes rigorously.
- Generalize the outcomes, not constrained in the common situations.
- An interesting journey, get a grasp of Advanced Mathematics.

How to Learn it Well

Differences between Elementary Math and Advanced Math:

- We define concepts to clarify what we are talking about. $\to$ Certainty.
- We use what we are certain about to model real world problems. $\to$ Modeling.
- We use our intelligence & mathetical tools to generalize what we learn / prove. $\to$ Thinking.
- " I'm good at math in High School" $\not\Leftrightarrow$ "I learn Vv186 well"

> *Please forget everything you have learned in school; you haven't learned it.*
>
> *Please keep in mind everywhere the corresponding portions of your school work; you haven't actually forgotten them.*

Edmund Landau, Foundations of Analysis, 1929

About Vv186

- Don't be frightened by "Honor" in the course title. Remember 156 is also an honor course.
- As you may be told by many, 186 focuses more on proofs and rigorous concepts, not calculations.
- It is not hard to get a good grade in 186, remember to attend lectures on time and finish assignments by yourself.
- Taking Notes is a good habit, personally.
- If you are worried about exams, make sure you finish assignments, sample exams and quizes.

How to Learn it Well (cont.)

- Attend every lecture and try to stay awake! During the lecture, focus on the general idea rather than getting stuck in technical details.
- Try to preview the slides before the lecture. Also review the course slides at least once a week. Focus on both general ideas and technical details this time.
- Do your assignment by your own and never depend too much on your teammate (Even though he/she has done every question perfectly, you'd better try to solve them independently first.).
- Discuss general questions regarding lectures with your friends.
- Come to recitation class (RC) to review knowledge and do some exercises.
- Come to office hours (OH) to ask questions, or simply have a chat with us.
- Many students find themselves struggle in learning VV186 at the beginning. We totally understand that! Don't worry too much, you can always turn to your Prof. & TAs.
- Be professional and respect TAs' privacy. (i.e. Use emails/piazza/OH rather than WeChat/QQ to contact us)

Continuity in Intervals

If, however, a function f is “well-behaved” enough such that we can use the same $\delta > 0$ for all $x \in I$, i.e., if in fact

$$\underset{\varepsilon>0}{\forall}\ \underset{\delta>0}{\exists}\ \underset{x\in I}{\forall}\ \underset{y\in I}{\forall} |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

we would want to give the function a special attribute.

2.5.22. Definition. Let $I \subset \mathbb{R}$ be an interval and $f : \Omega \to \mathbb{R}$ a function with $I \subset \Omega$. Then f is called ***uniformly continuous on I*** if and only if

$$\underset{\varepsilon>0}{\forall}\ \underset{\delta>0}{\exists}\ \underset{x,y\in I}{\forall} |x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon.$$

Clearly, if f is uniformly continuous on I, then f is also continuous on I.

Uniform Continuity on Closed Intervals

2.5.23. Theorem. Let $f\colon \Omega \to \mathbb{R}$ a function with $I = [a, b] \subset \Omega$. If f is continuous on $[a, b]$ then f is also uniformly continuous on $[a, b]$.

Proof.
Suppose that f is not uniformly continuous on I. Then

$$\underset{\varepsilon>0}{\exists}\ \underset{\delta>0}{\forall}\ \underset{x,y\in I}{\exists} \quad |x-y|<\delta \quad \wedge \quad |f(x)-f(y)| \geq \varepsilon.$$

Denote this ε by ε_0. Then for each $\delta = 1/n$ there exist numbers $x_n, y_n \in I$ such that

$$|x_n - y_n| < \frac{1}{n} \quad \wedge \quad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

We obtain two sequences (x_n) and (y_n) that are bounded (why?). By the theorem of Bolzano-Weierstraß (x_n) has a convergent subsequences (x_{n_k}), say with limit ξ. Consider now the sequence (y_{n_k}). It is also bounded and hence has a convergent subsequence $(y_{n_{k_j}})$, say with limit η.

Comments from Students

Comments from Former Students

1. “高中接觸的微積分都是高數內容，直到上完分析後才真正第二次愛上數學，在嚴謹的分享下有時卻藏著大膽的假設和論證。(A-)”
2. “Prof很好，Ta是神仙呜呜呜呜呜”
3. “数分是一个很值得沉下心认真学习的课，只要愿意认认真真学习，不断积累沉淀，可以拿到一个不错的分数！（A）”
4. “不知道我会不会是比较负面的回答。可能怪我自己学习方法和态度的问题。不只是186，很多JI课程我都是那种临时抱佛脚的状态。尽管最后成绩还算可以，但是对我来说，我其实并没有学到很多知识，基本都是考试前能对这门课的整个知识点有个大概的掌握，但是到了下学期，我对于这些知识会忘记很多。可能是我自己的学习问题，不过个人对于186的学习经历还挺负面的（A-）”
5. “186是一个逻辑清晰，难度极大，培养数学严格性和训练数学抽象思维的好课，虽只拿了A-但很值”
6. “186巨有意思思维量比很多课大很多然后蛮喜欢prof以及ta的讲课方式”

Comments from Students (cont.)

7. “感觉是一个其实比较flexible的课程？就其实大家如果做做sample其实也可以获得比较好的分数。但是如果对数学感兴趣，希望深入的理解课程的内容，更可以有意想不到的收获 ”
8. “喜欢抽象与逻辑世界的珍惜186吧！之后不会有这样严谨的、重视分析与证明的课程了。186是一门很基础的分析课程，苦日子还在后头 hss的魅力不仅在课堂，更在他的办公室里（A+）”
9. “感觉是个对思维能力要求很高的课程，但是也不强求，只要好好学还能拿个不错的分数。不过由于是进大学第一门课，在难度不低的同时还要适应JI的教学方式挺有挑战的。(A-, 但是第一次midq1都没考到)”
10. “尽可能搞懂每个定理怎么来的，为什么要讲，如果只是考前短时记忆定理内容，亲测会有很多翻车的时候；作业就算不做也要知道下队友怎么做的，有些你跳过的题很有可能变成考试的一部分。（A-，除了mid1比较简单临时看考的好，剩下两次都白给；后期听课作业都投入的有点少）”
11. **谨慎选课，不要跟风。选了要好好听课，写作业。（在后一学期跳车255了）**

Comments from Students (cont.)

12. 我特别喜欢火叔叔，真的能感受到火叔叔浑身散发着幽默以及对学生和教学的热情。他教授的不仅是数学知识，更是数学背后的思维和哲理。祝愿加入186的同学们能亲身去体会火叔叔的魅力！

13. 我特别喜欢horst的课，horst讲课往往做过充足的准备，使你在一秒进入状态的同时收获良多。同时horst考试试卷也安排的非常合理，区分度适中，难度适中。同时你还会认识数学小组的同学（往往是巨佬），还可以多认识朋友，让我对186课程十分满意。

14. 186是我上大学后学的的第一门数学课。在这门课中我学到了许多数学分析的基本思想。Horst在教授抽象的概念的时候经常会举一些形象的例子帮助同学们理解。很感谢horst老师给我们带来如此精彩，清晰，易懂的课程。

15. 有高等数学基础可以不用担心直接选 没有基础的话可能会比较吃力 但186实用性大于156

16. 186是一门很有深度的课程。在这门课里你可以从初等数学像高等数学逐步过渡。虽然课程的内容并不简单，颇具挑战，但是在这门课中打下的良好数学基础会为今后学习做一个很好的铺垫。更重要的是你会在这门课里遇到非常优异的同学，与他们一起共同进步。

Comments from Students (cont.)

17. 火叔叔很善良，大家可以从他那里学到很多知识。他会手把手教你积分，怎么微分，并且更重要的是，他会教会你什么是真正的数学。祝你享受在VV186的旅程！
18. 火叔叔准备了大量的定理推导，深入浅出的讲解让人瞬间爱上这门课。学完这门课后你会惊叹于和其他的课程相比，186是多么的与众不同。
19. 186感觉还是一门非常基础的并且注重概念的数学课，学起来非常自由，hss讲的很好，ta和ppt也肯定会非常nice（虽然学完之后感觉可能忘了不少，但是锻炼思维的过程同样也是非常可贵的）
20. 课很好， prof人很好， ta也很nice~ 但学不学得好还是看自己的努力程度。个人觉得不要把高中学数学的思路平移到大学，多想想数学的本质，好好体会下数分的乐趣。这门课很重要，因为285,286的内容很多都基于此展开~ 不要怕难，整个学期难度甚至有点递减的感觉，so 踏踏实实学下去一定会有好结果滴！

Comments from Students (cont.)

21. Horst 是一个非常慈祥的教授，数批必选。个人感觉186比起国内的高数体系而言，整个学习曲线比较丝滑，比如说级数在很前面的位置就介绍了，方便后面很多概念的建构和理解。就是建议自学积分等计算部分，相信TA们，也可以找点题做。另外，186的作业真的超好！做得很爽，能学到很多东西（虽然不一定对考试有用，但数批狂喜）。（A+）
22. 喜欢 Horst 的数学体系，比如用Cauchy序列构造实数就很有意思。RC质量很高，考试偏向于Sample Exam。(A+)

3. More...

Coursework

- There will be weekly coursework (assignments) throughout the term.
- You will be randomly assigned into ***assignment groups*** of three students; you are expected to collaborate within each group and hand in a single, common solution paper to each coursework.
- Each group must achieve 60% of the total coursework points by the end of the term in order to obtain a passing grade for the course. However, the assignment points have no effect on the course grade.
- Please hand in your coursework on time, by the date given on each set of course work. Late work will not be accepted unless you come to me personally and I find your explanation for the lateness acceptable.
- You can be deducted up to 10% of the awarded marks for an assignment if you fail to write neatly and legibly.
- Further details can be found in the course description.

Assignments and Grading (cont.)

Grading Policy

- The grade will be composed of the course work and the exams as follows:
  - Participation: 5 points
  - Course Outcome Quizzes: 5 points
  - Midterm exam: 45 points
  - Final exam: 45 points
- The actual grading scale will usually be based on the top approximately 6-12% of students receiving a grade of A+, with the following grades determined by (mostly) fixed point increments.
- Apart from this normalization, the grade distribution is up to you! If (for example) all students obtain many points in the exams, I am happy to see everyone receive a grade of A. Students are primarily evaluated with respect to a fixed point scale, not with respect to each other.

More...

Feel free to reach me out if you have any questions about the course.

Thank you!